James Rosado
Engineering Analysis
Summer 2015
Expository Paper

A Study of "Fundamental Solutions for Some Partial Differential Operators From Fluid Dynamics and Statistical Physics" (by Jorge Aarao)

Part 1: Introduction. The paper begins with a review of some of the fundamental nomenclature of PDE's and types of problems being solved. First, L is an operator that is used frequently in PDE's and represents a series of functions, operations being acted on a given variable or other function. For instance in the article, one of the PDE's studied has an operator L of the form,

$$L[u] = -\Delta u = -(u_{xx} + u_{yy} + u_{zz})$$

In other words, the operator is to take the second derivative with respect to each variable, take the sum, and then negate it. We could make up our own such as $L = \frac{\partial}{\partial x \partial y \partial z} + xy$, so it would look like the following,

$$L[u] = \frac{\partial u}{\partial x \partial y \partial z} + xy$$

The author studies PDE's of the form,

$$L[u(w)] = f(w)$$

The other uses the variable 'w' because later on we will be extending the solutions and PDE's variables on multiple dimensions; therefore,

$$L[u(\boldsymbol{w})] = f(\boldsymbol{w}) \rightarrow L[u(x,y,z)] = f(x,y,z)$$

At any rate, the solution to the differential equation, $L[u] = f$ is of the form,

$$u(w) = \int_{\mathbb{R}^n} f(w')G(w,w')dw'$$

The notation $\mathbb{R}^n$ means evaluating the integral over *n*-dimensional space; hence, $\mathbb{R}^1 =$ on real line, $\mathbb{R}^2 =$ on a plane, or $\mathbb{R}^3 =$ in a space and so on. The notation he uses is analogous to the notation utilized in Haberman's "Applied Differential Equation:"

Aarao	Haberman
$u(w) = \int_{\mathbb{R}^n} f(w')G(w,w')dw'$	$u(x) = \int_a^b f(x_0)G(x,x_0)dx_0$

Essentially, we are finding the corresponding Green's function to the given non-homogeneous differential equation, this form a fundamental solution to the original differential equation. In Aarao's article he explains in depth the following examples that appear frequently in fluid dynamics and physics,

1. $L[u] = -\Delta u = -(u_{xx} + u_{yy} + u_{zz})$
2. $L[u] = xu_y - u_{xx}$
3. $L[u] = xu_y - yu_x - u_{xx}$
4. $L[u] = kxu_x - kyu_y - \Delta u$
5. $L[u] = -2kyu_z - \Delta u$

Part 2: The Method. For each equation above we are trying to find *u* such that $L[u] = f$. We will first examine the evolution equation,

$$v_t + L[v] = 0$$

Where, $v(w,0) = f(w)$. Here Aarao makes the following definition,

$$u(w) = \int_0^\infty v(w,t)\,dt$$

If we substitute the definition above into our equation $L[u] = f$, we get the following,

$$L[u] = L\left[\int_0^\infty v(w,t)\,dt\right] = \int_0^\infty L[v]\,dt$$

But recall that, $v_t + L[v] = 0$ ➔ $L[v] = -v_t$; therefore,

$$L[u] = \int_0^{\infty} L[v]\, dt = -\int_0^{\infty} v_t\, dt = -v(t)|_{t=0}^{t=\infty} = v(0) - v(\infty)$$

Let us examine the last statement above, $v(0)$ is the initial behavior of our function *v* at time t = 0, but we already established that $v(w, 0) = f(w)$, thus $v(0) = f$. The second term is the steady state behavior of the heat, but we know that as time continues the heat will be dissipated; therefore, $v(\infty) = 0$. We have established that our evolution equation will satisfy the original differential equation $L[u] = f$. In Aarao's article, he suggest that the solution to $v(w, t)$ we be of the form,

$$v(w, t) = \int_{\mathbb{R}^n} f(w')H(w, w'; t)\, dw'$$

The function $H(w, w'; t)$ is called the "heat kernel" that corresponds to the given operator *L*. The means that there is not one singular solution all *L* operator's but we need to solve for this function and it will depend on the function *f* and the boundary conditions.—in a way it is analogous to the finding the corresponding the Green's function of the differential equation, which Aarao utilizes next. Aarao makes the following statement on finding the heat kernel,

$$u(w) = \int_0^{\infty} v(w, t)dt = \int_0^{\infty} \left(\int_{\mathbb{R}^n} f(w')H(w, w'; t)\, dw' \right) dt$$

But notice that the inner integral above can be moved outside the outer integral because the inner integral depends on *w'* while the outer one only depends on *t*, we get,

$$u(w) = \int_{\mathbb{R}^n} f(w') \left(\int_0^{\infty} H(w, w'; t)\, dt \right) dw'$$

Notice that the inner integral when evaluated will only be a function on w, w'. We will make this inner integral our Green's function,

$$u(w) = \int_{\mathbb{R}^n} f(w')G(w, w')\, dw'$$

Aarao make the following claims, first that since *G* the Green's function is obtained from the heat kernel *H*, the Green's function and heat kernel are also solutions to the same type of differential operator,

$$H_t + L[H] = 0$$

This makes sense because,

$$v_t + L[v] = 0 \rightarrow \left(\frac{\partial}{\partial t} + L\right) v(w,t) = 0$$

If we substitute $v(w,t) = \int_{\mathbb{R}^n} f(w')H(w,w';t)\,dw'$ we get,

$$\left(\frac{\partial}{\partial t} + L\right) \int_{\mathbb{R}^n} f(w')H(w,w';t)\,dw' = 0$$

We can move the differential operator inside the integral and get,

$$\int_{\mathbb{R}^n} f(w') \left(\frac{\partial}{\partial t} + L\right) H(w,w';t)\,dw' = 0$$

Immediately notice that $f(w') \neq 0$ and the only way for the integral to equal 0 is for the differential operator acting on H is 0 or,

$$\left(\frac{\partial}{\partial t} + L\right) H(w,w';t) \rightarrow H_t + L[H] = 0$$

Secondly recall, that $v(w,0) = f(w)$ and $v(w,t) = \int_{\mathbb{R}^n} f(w')H(w,w';t)\,dw'$. In the integral if we let t = 0 we get,

$$f(w) = v(w,0) = \int_{\mathbb{R}^n} f(w')H(w,w';0)\,dw'$$

Or more directly,

$$f(w) = \int_{\mathbb{R}^n} f(w')H(w,w';0)\,dw'$$

Notice that this is analogous to the behavior of the delta function centered at a point, recall the property,

$$\int_{-\infty}^{\infty} f(w')\delta(w'-w)dw' = f(w)$$

This means that $H(w,w';0) = \delta(w'-w)$. Therefore, we have established a new differential equation with conditions,

- $H_t + L[H] = 0$

- $H(w,w';0)=\delta(w'-w)$

Therefore, we need to find *H* because then we can find the Green's function *G* and then we can find the solution *u*. From Haberman [pp. 520-521] The solution to the differential equation $\frac{\partial G}{\partial t}=k\nabla^2 G$ for a one dimensional rod is $G(x,t;x_0,t_0)=\frac{1}{\sqrt{4\pi k(t-t_0)}}e^{-\frac{(x-x_0)^2}{4k(t-t_0)}}$. For our problem of the heat kernel *H* we will let *k* = 1 and $t_0=0$. We will get,

$$G(x,t;x_0)=\frac{1}{\sqrt{4\pi t}}e^{-\frac{(x-x_0)^2}{4t}}$$

We will switch back to our notation and let $w=x$ and $w'=x_0$ and we get,

$$H(w,w';t)=\frac{1}{\sqrt{4\pi t}}e^{-\frac{(w-w\prime)^2}{4t}}$$

This makes sense considering that our heat kernel is analogous to the Green's function. But, this solution is for a one-dimensional rod, for *n* dimensions our heat kernel becomes,

$$H(w,w';t)=\frac{1}{(4\pi t)^{\frac{n}{2}}}e^{-\frac{(w-w\prime)^2}{4t}}$$

The author makes the following assertion; we can represent this heat kernel as the following function acting on matrices and vectors in $\mathbb{R}^n$.

$$H(w,w';t)=\phi(t)e^{-\frac{1}{2}\langle Ah,h\rangle}$$

Where, *A* is a matrix of coefficients that depend on the variable *t* time variable and it is 2n by 2n in size,

$$A=\begin{bmatrix}a(t) & b(t)\\ c(t) & d(t)\end{bmatrix}$$

And *h* are the column vectors representing the variables:

$$\begin{bmatrix}w\\ w'\end{bmatrix}=\begin{bmatrix}(x,y,z)\\ (x',y',z')\end{bmatrix}$$

The notation $\langle Ah,h\rangle$ represents the dot product:

$$Ah\cdot h=\begin{bmatrix}a(t) & b(t)\\ c(t) & d(t)\end{bmatrix}\begin{bmatrix}w\\ w'\end{bmatrix}\cdot\begin{bmatrix}w\\ w'\end{bmatrix}=\begin{bmatrix}aw+bw'\\ cw+dw'\end{bmatrix}\cdot\begin{bmatrix}w\\ w'\end{bmatrix}=aw^2+bw'w+cww'+dw'^2$$

Part 3: Calculation.

The author utilizes the following property involving the dot product, $\langle h,h_1\rangle=h^Th_1=h_1^Th$ and the symbol $\dot{\phi}$ represents the derivative with respect to time. Remember,

$$H(w,w';t)=\phi e^{-\frac{1}{2}\langle Ah,h\rangle}$$

Where, *A* is a symmetric non-negative matrix function on *t*. Now we will substitute this into our differential equation: $H_t+L[H]=0$.

$$H_t = \frac{\partial H}{\partial t} = \frac{\partial}{\partial t}\left(\phi e^{-\frac{1}{2}\langle Ah,h\rangle}\right) = e^{-\frac{1}{2}\langle Ah,h\rangle}\frac{\partial \phi}{\partial t} + \phi\frac{\partial}{\partial t}\left(e^{-\frac{1}{2}\langle Ah,h\rangle}\right)$$

$$H_t = \dot{\phi}e^{-\frac{1}{2}\langle Ah,h\rangle} - \phi\left(\frac{1}{2}\right)\langle \dot{A}h,h\rangle e^{-\frac{1}{2}\langle Ah,h\rangle}$$

Notice we can express the derivative above in terms of H,

$$H_t = \frac{\dot{\phi}}{\phi}\phi e^{-\frac{1}{2}\langle Ah,h\rangle} - \left(\frac{1}{2}\right)\langle \dot{A}h,h\rangle \phi e^{-\frac{1}{2}\langle Ah,h\rangle}$$

$$H_t = \left(\frac{\dot{\phi}}{\phi} - \left(\frac{1}{2}\right)\langle \dot{A}h,h\rangle\right)H$$

We take the derivative with respect to the x variable and get,

$$H_x = \frac{\partial H}{\partial x} = \frac{\partial}{\partial x}\ \phi e^{-\frac{1}{2}\langle Ah,h\rangle}$$

$$H_x = \phi\frac{\partial}{\partial x}e^{-\frac{1}{2}\langle Ah,h\rangle} = \phi e^{-\frac{1}{2}\langle Ah,h\rangle}\left(-\frac{1}{2}\frac{\partial}{\partial x}\langle Ah,h\rangle\right) = H\left(-\frac{1}{2}\frac{\partial}{\partial x}\langle Ah,h\rangle\right)$$

Thus,

$$H_x = H\left(-\frac{1}{2}\frac{\partial}{\partial x}\langle Ah,h\rangle\right)$$

But what is $\frac{\partial}{\partial x}\langle Ah,h\rangle$? So we will think of this as our usual dot-product notation. Notice that $h = (x,y,z,\dots)$ and if we take the derivative with respect to x we get $h_x = (1,0,0,0,\dots) = e_1$,

$$\begin{aligned}\frac{\partial}{\partial x}\langle Ah,h\rangle &= \frac{\partial}{\partial x}A(t)h\cdot h\\ &= A(t)\frac{\partial h}{\partial x}\cdot h + A(t)h\cdot\frac{\partial h}{\partial x}\\ &= Ae_1\cdot h + Ah\cdot e_1\\ &= 2Ah\cdot e_1 = 2\langle Ah,e_1\rangle\end{aligned}$$

Hence,

$$H_x = H\left(-\frac{1}{2}2\langle Ah,e_1\rangle\right) = -\langle Ah,e_1\rangle H$$

We can extend this to more than one dimension and get similar results,

$$H_y = H\left(-\frac{1}{2}2\langle Ah,e_1\rangle\right) = -\langle Ah,e_2\rangle H$$

$$H_z = H\left(-\frac{1}{2}2\langle Ah,e_1\rangle\right) = -\langle Ah,e_3\rangle H$$

The second derivative has a more interesting result,

$$H_{xx} = \frac{\partial H_x}{\partial x} = -\frac{\partial}{\partial x}(\langle Ah,e_1\rangle H)$$

$$= -\langle Ah, e_1\rangle H_x - H\frac{\partial}{\partial x}\langle Ah, e_1\rangle$$
$$= \langle Ah, e_1\rangle^2 H - H\langle Ah_x, e_1\rangle$$
$$= \langle Ah, e_1\rangle^2 H - H\langle Ae_1, e_1\rangle$$

But what is this last statement above? Recall, that A is a matrix of 2n by 2n dimension. So as an example, let,

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

So,

$$Ae_1 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a_{11}\cdot 1 + a_{12}\cdot 0 \\ a_{21}\cdot 1 + a_{22}\cdot 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}$$

Now when we evaluate the dot product we get,

$$\langle Ae_1, e_1\rangle = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}\cdot\begin{bmatrix} 1 \\ 0 \end{bmatrix} = a_{11}\cdot 1 + a_{21}\cdot 0 = a_{11}$$

Therefore, $\langle Ae_k, e_k\rangle$ will filter out functions along the diagonal of A. Hence, $H_{xx} = \langle Ah, e_1\rangle^2 H - Ha_{11}$. Let us now examine the first term for H_{xx}: $\langle Ah, e_1\rangle^2$; recall the dot product property form before,

$$\langle Ah, e_1\rangle^2 = \langle Ah, e_1\rangle\langle Ah, e_1\rangle$$
$$= ((Ah)^T e_1)((Ah)^T e_1)$$
$$= (h^T A e_1)(e_1^T Ah)$$
$$= h^T A P_1 Ah$$

Where, $P_1 = e_1 e_1^T$, this is a matrix of zeroes except at the position (1,1) there is a 1. Notice that A, P_1 are both symmetric matrices meaning that $A = A^T$ and $P_1 = P_1^T$. So we can consider the last statement as another dot product,

$$\langle Ah, e_1\rangle^2 = \langle AP_1 Ah, h\rangle$$

Therefore, our second derivative is,

$$H_{xx} = \langle AP_1 Ah, h\rangle H - Ha_{11}$$
$$H_{yy} = \langle AP_2 Ah, h\rangle H - Ha_{22}$$
$$H_{zz} = \langle AP_3 Ah, h\rangle H - Ha_{33}$$

So what does this have to do with finding our solution to the differential equation $L[u] = f$. Recall that we are beginning with the evolution equation $v_t + L[v] = 0$. This equation takes our non-homogeneous equation and "makes" it homogeneous; therefore, if we can find the solutions to the evolution equation then we can find the solutions to our original non-homogeneous equation. Utilizing the evolution equation we can find a solution by finding the heat kernel H, this is a type of Green's function. What Aarao has pointed out, is that since we were able to find

the solution to the heat kernel in one dimension, is it possible to find it for multiple dimensions? The answer is yes, it involves linear algebra manipulations.

Part 4: "Matrix System"

We will consider the two-dimensional case for simplicity, since for higher dimensional analysis the work will be the same. The author utilizes this definition for the operator *L*,

$$L = (k_1x + k_2y)\partial_x + (k_3x + k_4y)\partial_y - \left(\tau_1\partial_{xx} + \tau_2\partial_{yy}\right) + \kappa$$

To implement this operator in our differential equation $H_t + L[H] = 0$, recall the derivatives we established earlier,

- $H_t = \left(\frac{\dot{\phi}}{\phi} - \left(\frac{1}{2}\right)\langle \dot{A}h, h\rangle\right)H$
- $H_x = -\langle Ah, e_1\rangle H$
- $H_y = -\langle Ah, e_2\rangle H$
- $H_{xx} = \langle AP_1Ah, h\rangle H - Ha_{11}$
- $H_{yy} = \langle AP_2Ah, h\rangle H - Ha_{22}$

Notice that all of the derivatives above have a factor of *H*, hence by division it will divide out of the differential equation.

$$H_t + L[H] = \left(\frac{\dot{\phi}}{\phi} - \left(\frac{1}{2}\right)\langle \dot{A}h, h\rangle\right)H + (k_1x + k_2y)H_x + (k_3x + k_4y)H_y - \left(\tau_1H_{xx} + \tau_2H_{yy}\right) + \kappa$$

$$= \left(\frac{\dot{\phi}}{\phi} - \left(\frac{1}{2}\right)\langle \dot{A}h, h\rangle\right)H - (k_1x + k_2y)\langle Ah, e_1\rangle H - (k_3x + k_4y)\langle Ah, e_2\rangle H - \left(\tau_1(\langle AP_1Ah, h\rangle H - Ha_{11}) + \tau_2(\langle AP_2Ah, h\rangle H - Ha_{22})\right) + \kappa$$

This simplifies to,

$$= \frac{\dot{\phi}}{\phi} - \left(\frac{1}{2}\right)\langle \dot{A}h, h\rangle - (k_1x + k_2y)\langle Ah, e_1\rangle - (k_3x + k_4y)\langle Ah, e_2\rangle - \left(\tau_1(\langle AP_1Ah, h\rangle - a_{11}) + \tau_2(\langle AP_2Ah, h\rangle - a_{22})\right) + \kappa$$

If we rearrange the terms, we get,

$$\frac{\dot{\phi}}{\phi} - \frac{1}{2}\langle \dot{A}h, h\rangle - \langle Ah, Sh\rangle + (\tau_1a_{11} + \tau_2a_{22}) - \langle APAh, h\rangle + \kappa = 0$$

And let,

$$S = \begin{bmatrix}\sigma & 0\\ 0 & 0\end{bmatrix},\ P = \begin{bmatrix}p & 0\\ 0 & 0\end{bmatrix},\ \sigma = \begin{bmatrix}k_1 & k_2\\ k_3 & k_4\end{bmatrix},\ p = \begin{bmatrix}\tau_1 & 0\\ 0 & \tau_2\end{bmatrix}$$

From here we can determine the time-dependent coefficients of the matrix $\dot{A}$, remember $b(t) = c(t)$ because the matrix is symmetric,

$$\dot{A} = \begin{bmatrix} \dot{a} & \dot{b} \\ \dot{b} & \dot{d} \end{bmatrix} \Rightarrow \begin{cases} \dot{a}(t) = -2apa - (a\sigma + \sigma^T a) \\ \dot{b}(t) = -2apb - \sigma^T b \\ \dot{d}(t) = -2cpb = -2b^T pb \\ \dot{\phi}(t) = -(\tau_1 a_{11} + \tau_2 a_{22})\phi - \kappa\phi = -trace(pa)\phi - \kappa\phi \end{cases}$$

For the last equation the trace of a matrix is the sum of the elements along the main diagonal. Notice we have arrived at a system of four equations with four unknowns and we can solve for the unknown quantities.

Part 5: An Example.

Recall the operator, $L[u] = -2kyu_z - \Delta u$, in the study of Poiseuille Flow the following differential equation arises

$$\frac{\partial C}{\partial t} + (1 - r^2)\frac{\partial C}{\partial z} = D\left(\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} + \frac{1}{r^2}\frac{\partial^2 C}{\partial \theta^2} + \frac{\partial^2 C}{\partial z^2}\right)$$

The constant 'D', known as the diffusivity, depends on the application of the fluid flow i.e. blood flow, sewage dispersion..and it is equation to $\frac{1}{Pe}$, where *Pe* is the Peclet number. The function 'C' is the concentration of passing fluid. At t = 0 the concentration is not changing with respect to time thus $\frac{\partial C}{\partial t} = 0$ and the right hand side of the equation is equivalent to take the Laplacian operator Δ with respect to the concentration function, we will let D = 1 for simplicity and we get,

$$(1 - r^2)C_z = \Delta C \rightarrow (constant)C_z - \Delta C = 0$$

This problem is analogous to the problem $L[u] = -2kyu_z - \Delta u = 0$. Therefore we will solve this general form of the fluid flow problem. Recall that our goal is to find the heat kernel because from there we can find our solution to the original differential equation. We need to find the coefficients of the matrix $\dot{A}$, it turns out with some analysis and linear algebra we get the following result,

$$A = \frac{1}{2t\left(1 + \frac{k^2t^2}{3}\right)}\begin{bmatrix} 1 + \frac{k^2t^2}{3} & 0 & 0 \\ 0 & 1 + \frac{4k^2t^2}{3} & kt \\ 0 & kt & 1 \end{bmatrix}$$

Notice that his matrix is symmetric, $A = A^T$ and it is non-negative. Also we can obtain the eigenvalues for our system of equations,

$$\lambda = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 2kt & -1 \end{bmatrix}$$

The corresponding eigen-function ϕ will be,

$$\phi(t) = \frac{1}{8\pi^{\frac{3}{2}} t^{\frac{3}{2}} \left(1 + \frac{k^2 t^2}{3}\right)^{\frac{1}{2}}}$$

Thus our solution to the heat kernel is,

$$H(w, w'; t) = \frac{1}{8\pi^{\frac{3}{2}} t^{\frac{3}{2}} \left(1 + \frac{k^2 t^2}{3}\right)^{\frac{1}{2}}} e^{-\frac{1}{2}\langle Ah, h\rangle}$$

Final Remarks. The methodology that Aarao presents uses linear algebra and basic calculus, and in some cases this is more direct than other methods of finding the heat kernels. Some issues that arise is whether the integrals that describe the Green's function converge, the author gives the example that when $L = -\Delta$ we get a Green's function that contains a logarithm in it, the integral of this does not converge on the domain $[0, \infty]$. Aarao also make an intuitive argument that our operators L must fit a "positivity" criteria,

$$\int_{\mathbb{R}^n} f(w) L[f] dw \geq 0$$

This makes sense since we what our integrals for the solution to converge, another way to think of this is that eh Euclidean norm on the solution to decay as time continues to ∞ or,

$$\frac{d}{dt} ||v||^2 \to 0$$

REFERENCES

[1] J. Aarao, *Fundamental Solutions for Some Partial Differential Operators from Fluid Dynamics and Statistical Physics*, SIAM, 49 (2015), pp. 303-314.

[2] M. Latini and A. J. Bernoff, *Transient anomalous diffusion in Poiseuille flow*, J. Fluid Mech., 441 (2001), pp. 399-411.

[3] R. Haberman, *Applied Differential Equation with Fourier Series and BVP,* 5th ed., Pearson (2013).